

|    |   |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |   | $T = \sqrt{\frac{2R}{GM}}$                                                                                                                                                                                                                                                                                                                                                                                            |
| 12 | D | <div>Heat gained by A = Heat lost by B</div> <div> <math>m_A c_A \Delta\theta_{A \rightarrow T}</math> <math>=</math> <math>m_B c_B \Delta\theta_{B \rightarrow T}</math> </div> <div> <math>m_A c_A (75)</math> <math>=</math> <math>m_B c_B (25)</math> </div> <div> <math>\frac{C_A}{C_B}</math> <math>=</math> <math>\frac{m_B (25)}{m_A (75)}</math> </div> <div> <math>=</math> <math>\frac{4}{3}</math> </div> |
| 13 | A | <div>Total number of moles before = Total number of moles after</div> <div> <math>P V_1 = P(V_1 + V_2)</math> </div>                                                                                                                                                                                                                                                                                                  |